Customer Sales & Churn Analysis Report

1. Project Overview

This project combines **sales transaction data** and **customer churn data** to uncover insights into customer behavior, product performance, regional sales, and retention risks.
The analysis was conducted using **Python (pandas, matplotlib, seaborn)** with a structured 7-day workflow covering data preparation, analysis, visualization, and reporting.

Objectives:

- Identify most valuable customers and calculate Customer Lifetime Value (CLV).
- Detect product bundling and cross-selling opportunities.
- Compare regional sales performance.
- Analyze seasonal sales trends.
- Understand churn drivers and propose retention strategies.
- Build a professional dashboard with 4–5 key visualizations.

2. Setup & Installation

Requirements:

- Python 3.9+
- Jupyter Notebook or VS Code
- Libraries: pandas, matplotlib, seaborn, plotly

Installation:

```
pip install pandas matplotlib seaborn plotly
```

Execution:

1. Place sales_data.csv and customer_churn.csv in the working directory.
2. Run the notebook/script (customer_analysis.ipynb or .py).
3. Outputs include tables, charts, and this report.

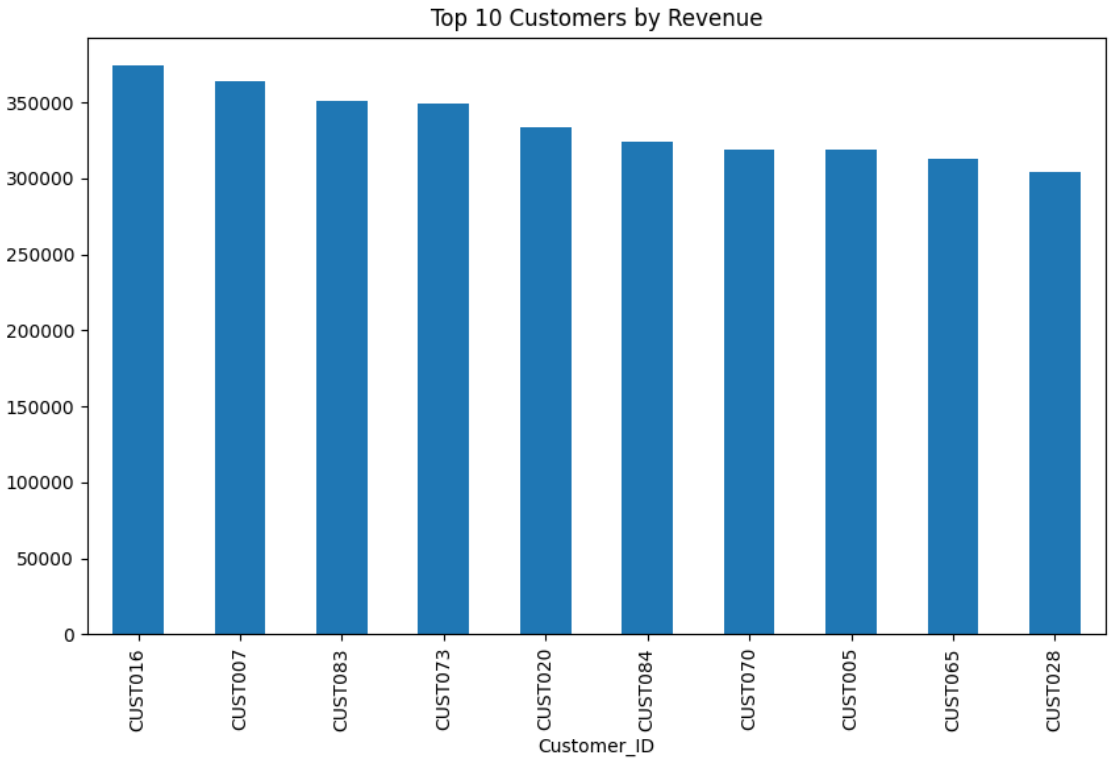
3. Code Structure

- **Day 1:** Data loading & exploration
- **Day 2:** Cleaning & preparation (date conversion, calculated fields)
- **Day 3:** Customer analysis (top customers, CLV, regional distribution)
- **Day 4:** Sales pattern analysis (monthly trends, best-selling products)

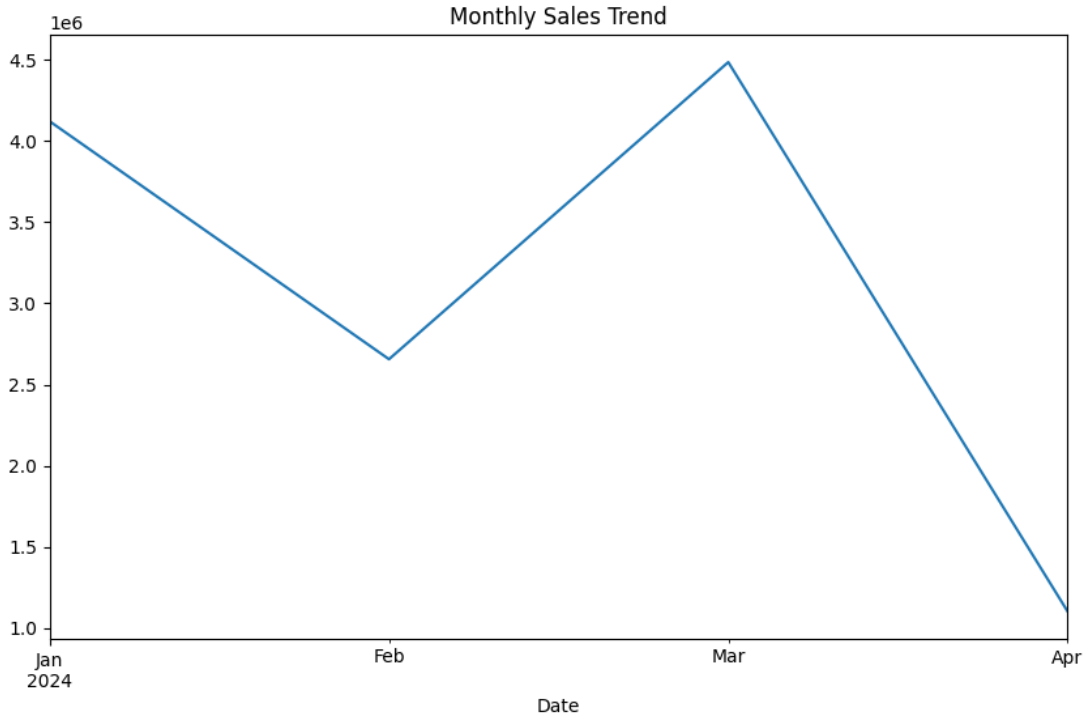
- **Day 5:** Advanced analysis (pivot tables, retention, cross-selling)
- **Day 6:** Dashboard creation (bar, line, stacked bar, pie, heatmap)
- **Day 7:** Report & insights

4. Key Visualizations

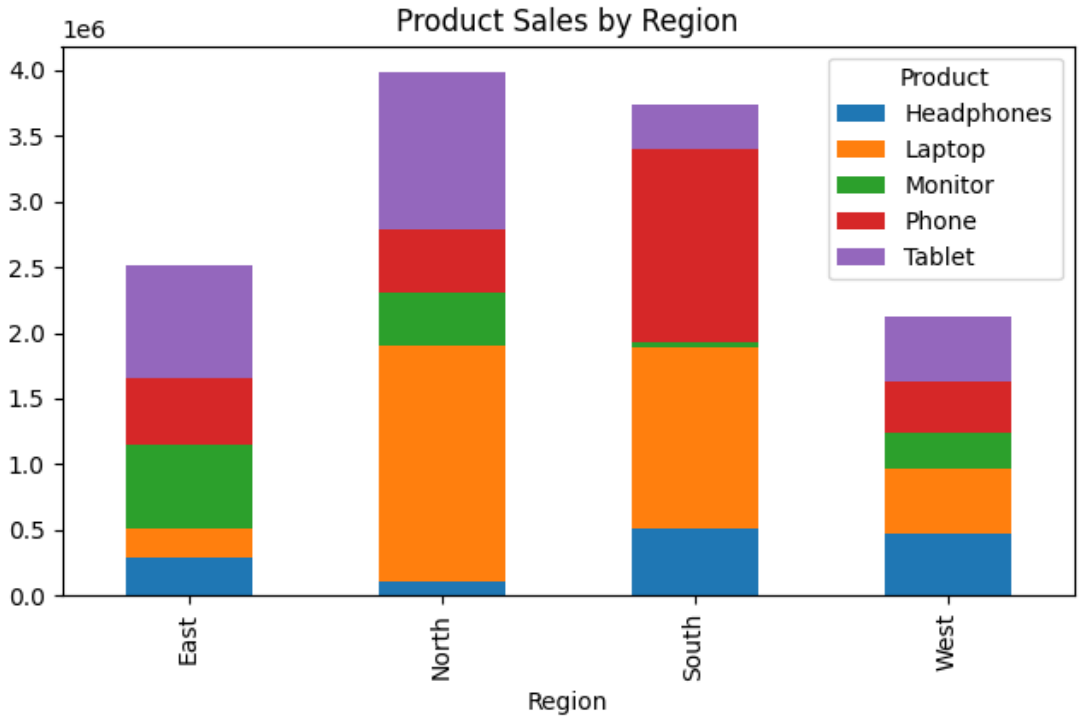
- Bar chart of top customers.



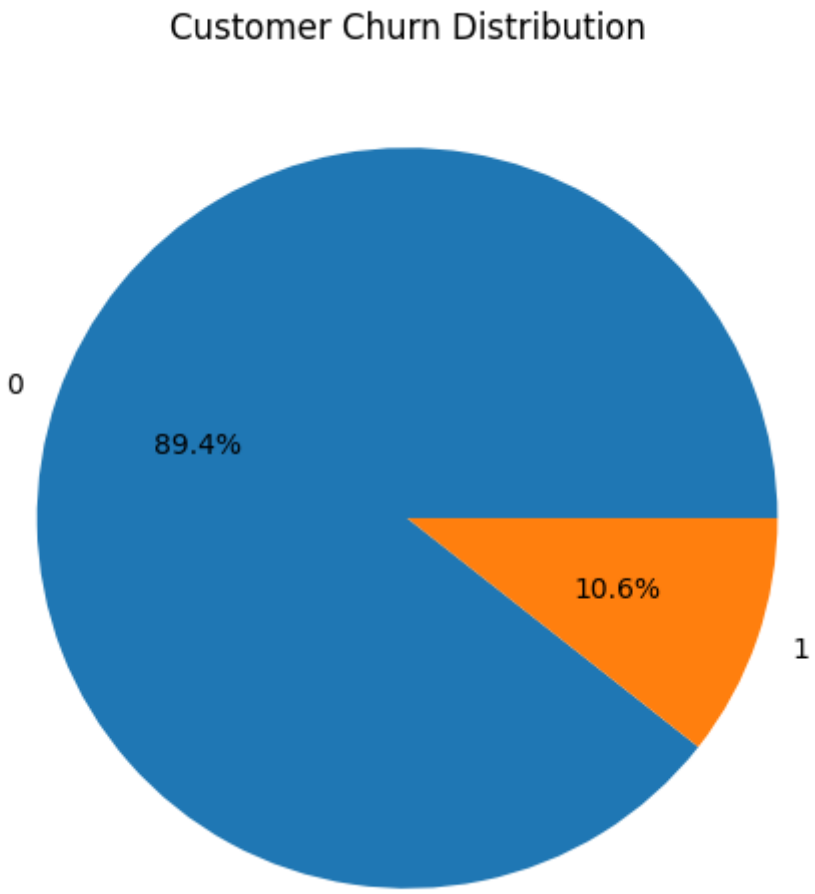
- Line chart of monthly sales trend.



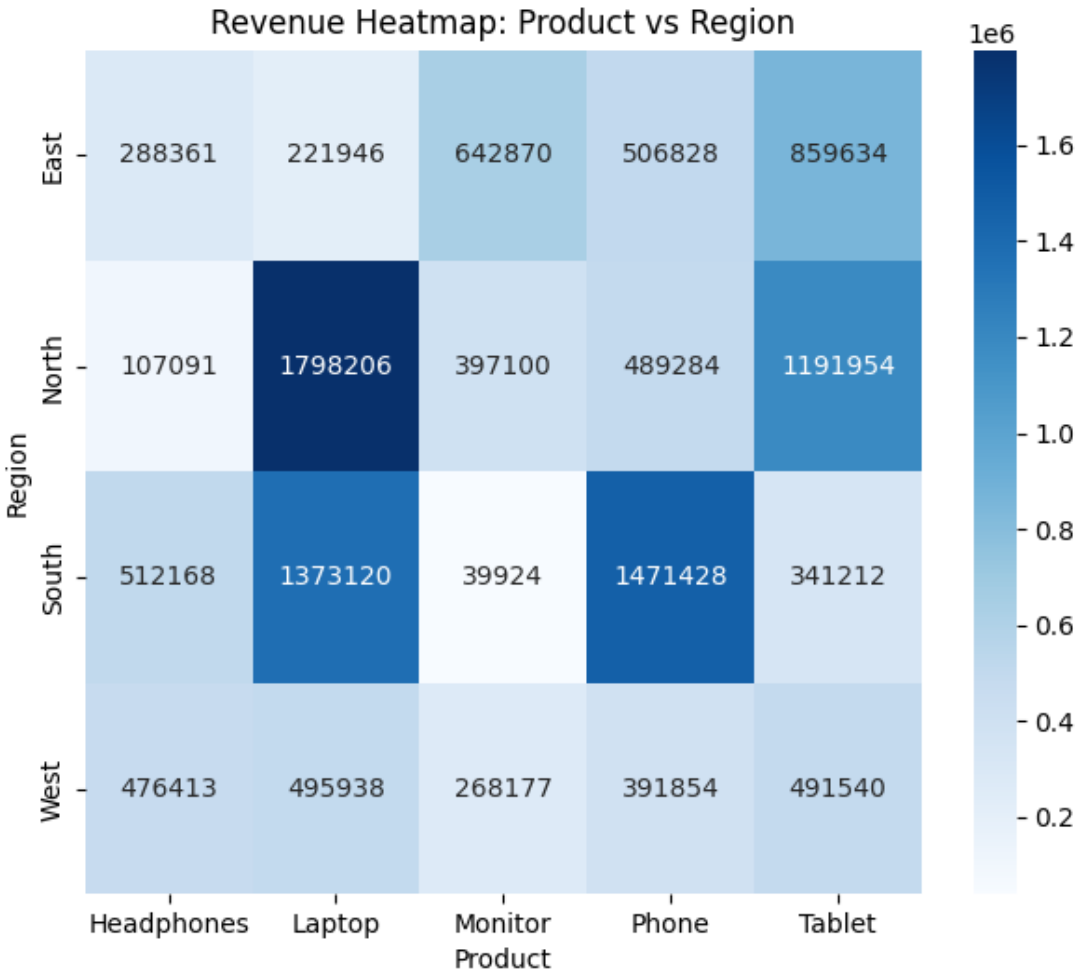
- Stacked bar chart of product sales by region.



- Pie chart of churn distribution.



- Heatmap of product vs region revenue.



5. Analysis Results

- **Most Valuable Customers:**
  - **CUST016** – $373,932 (Phones)
  - **CUST073** – $349,510 (Headphones)
  - Other high-value customers include CUST005, CUST007, CUST020.
- **Products that Sell Together:**
  - Laptop + Monitor (work setup bundle)
  - Phone + Headphones (mobile accessory bundle)
- **Regional Sales:**
  - North region leads in revenue (strong laptop & tablet demand).
  - South region follows (high phone sales).

    - East region lags behind.
- **Seasonal Trends:**
    - Peaks in January and March (new product launches).
    - February shows moderate sales, driven by monitors.
- **Customer Retention:**
- Month-to-month contracts have highest churn risk.
- Customers using Electronic Check churn more than those using credit cards/bank transfers.

6. Technical Requirements Met

- **pandas for data manipulation**: groupby, merge, pivot_table.
- **3+ aggregations**: customer revenue, regional totals, monthly sales.
- **Data merging**: sales data joined with churn data.
- **Pivot tables**: product vs region revenue.
- **Visualizations**: bar, line, stacked bar, pie, heatmap.

7. Executive Summary

- **Top Customer:** CUST016 – $373,932
- **Best Bundles:** Laptop + Monitor, Phone + Headphones
- **Highest Sales Region:** North
- **Seasonal Peaks:** January & March
- **Retention Strategy:** Target month-to-month customers, incentivize secure payment methods

Recommendations:

1. Launch loyalty programs for high-value customers.
2. Bundle laptops with monitors and phones with headphones.
3. Strengthen marketing in East region.
4. Offer discounts for long-term contracts to reduce churn